Elementary Problems
Divisibility
Solutions

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Basic Work

1. We have $n^2 - 1 = (n-1)(n+1)$. Since $n-1$ and $n+1$ are two consecutive even numbers, one of them must be a multiple of 4 and the other is divisible by 2. Therefore, their product is divisible by $4 \times 2 = 8$.

2. we have

$$\begin{aligned}(1234, 5678) &= (1234, 5678 - 1234 \times 4) \\ &= (1234, 742) = (1234 - 742, 742) \\ &= (492, 742) = (492, 742 - 492) \\ &= (492, 250) = (250 \times 2 - 492, 250) \\ &= (8, 250) = 2.\end{aligned}$$

3. We have
$$(a+b, a-b) = (a+b, (a-b) + (a+b)) = (a+b, 2a).$$
Since $(a, b) = 1$, we have $(a+b, a) = (b, a) = 1$. Thus,
$$(a+b, a-b) = (a+b, 2).$$
Therefore, the greatest common divisor is 1 if a and b have different parities, and is 2 if they have the same parity.

4. Firstly, we have
$$n^{120} = n^{15}(n^{35})^3 \equiv n^{15}(1)^3 = n^{15} \pmod{n^{35} - 1}.$$
This implies $(n^{120} - 1, n^{35} - 1) = (n^{15} - 1, n^{35} - 1)$. Similarly, we have
$$(n^{15} - 1, n^{35} - 1) = (n^{15} - 1, n^{35 - 15\times 2} - 1) = (n^{15} - 1, n^5 - 1) = n^5 - 1.$$
In general, we have $(n^a - 1, n^b - 1) = n^{(a,b)} - 1$.

5. It is the same as showing $(21n + 4, 14n + 3) = 1$. Indeed, we have
$$\begin{aligned}(21n + 4, 14n + 3) &= ((21n + 4) - (14n + 3), 14n + 3) = (7n + 1, 14n + 3) \\ &= (7n + 1, (14n + 3) - 2(7n + 1)) = (7n + 1, 1) = 1.\end{aligned}$$

6. Firstly, we have $(a, ab + cd) = (a, cd) = 1$ since $(a, c) = (a, d) = 1$.

   Secondly, we have $(c, ab + cd) = (c, ab) = 1$ since $(c, a) = (c, b) = 1$.

   These combine to give $(ac, ab + cd) = 1$.

7. We prove $(a_n, a_{n+1}) = 1$ by induction on n. The base case $n = 1$ is trivial.
   Assume $(a_k, a_{k+1}) = 1$ for some $k \geqslant 1$. Then

$$(a_{k+1}, a_{k+2}) = (a_{k+1}, 4a_{k+1} + a_k) = (a_{k+1}, 4a_{k+1} + a_k - 4a_{k+1}) = (a_{k+1}, a_k) = 1.$$

   Hence, the claim is true by induction.

8. We have

$$(a_n, a_{n+1}) = (n^2 + 20, (n+1)^2 + 20) = (n^2 + 20, 2n + 1) = (2n^2 + 40, 2n + 1)$$

   since $2n + 1$ is odd. Then

$$(a_n, a_{n+1}) = (-n + 40, 2n + 1) = (-2n + 80, 2n + 1) = (81, 2n + 1).$$

   This shows the greatest common divisor divides 81.

9. Suppose on the contrary that $(a_1 - 1)(a_2 - 2) \cdots (a_n - n)$ is odd. Then each $a_j - j$ is odd, and hence

$$(a_1 - 1) + (a_2 - 2) + \cdots + (a_n - n)$$

   is odd since there is an odd number of odd terms. However, we have

$$(a_1 - 1) + (a_2 - 2) + \cdots + (a_n - n) = (a_1 + a_2 + \cdots + a_n) - (1 + 2 + \cdots + n) = 0.$$

   This is a contradiction. Thus, $(a_1 - 1)(a_2 - 2) \cdots (a_n - n)$ is even.

10. Since $n^2 + 1 = (n - 1)(n + 1) + 2$, we have $n + 1 \mid 2$. It follows that $n + 1 = \pm 1, \pm 2$. Correspondingly, we get $n = -3, -2, 0, 1$. The steps are reversible so that all of these are solutions.

11. Since $2n + 6 = 2(n - 4) + 14$, we have $n - 4 \mid 14$. It follows that $n - 4 = \pm 1, \pm 2, \pm 7, \pm 14$. Correspondingly, we get the positive solutions $n = 2, 3, 5, 6, 11, 18$. The steps are reversible so that all of these are solutions.

12. From $3n - 1 \mid 2n + 2$, we have $3n - 1 \mid 3(2n + 2) = 2(3n - 1) + 8$. So $3n - 1 \mid 8$. It follows that $3n - 1 = \pm 1, \pm 2, \pm 4, \pm 8$. Correspondingly, we get the positive solutions $n = 1, 3$. We check that

$$3(1) - 1 \mid 2(1) + 2 \quad \text{and} \quad 3(3) - 1 \mid 2(3) + 2.$$

13. From $17 \mid 2m + 3n$, we have $17 \mid 13(2m + 3n)$. As

$$13(2m + 3n) = 26m + 39n = 9m + 5n + 17(m + 2n),$$

   this implies $17 \mid 9m + 5n$.

14. Suppose on the contrary that $\sqrt{2} = \dfrac{m}{n}$ for some positive integers m, n. WLOG we may assume $(m, n) = 1$. Squaring both sides, we get $2n^2 = m^2$. This shows m is even.

    Let $m = 2k$. Then $2n^2 = m^2 = 4k^2$. Hence, $n^2 = 2k^2$. This shows n is even. However, this contradicts $(m, n) = 1$ as m and n are both even. Thus, $\sqrt{2}$ is irrational.

15. Suppose on the contrary that $\sqrt{2} + \sqrt{3} = \dfrac{m}{n}$ for some positive integers m, n. Squaring both sides, we get $5 + 2\sqrt{6} = \dfrac{m^2}{n^2}$. This yields

    $$\sqrt{6} = \frac{m^2 - 5n^2}{2n^2}.$$

    As m, n are positive integers, this implies $\sqrt{6}$ is a rational number. However, it is stardard to show that $\sqrt{6}$ is irrational. This is a contradiction. Thus, $\sqrt{2} + \sqrt{3}$ is irrational.

16. For example, each of the n numbers

    $$(n+1)! + 2,\ (n+1)! + 3,\ \ldots,\ (n+1)! + (n+1)$$

    is composite since $(n+1)! + j$ is divisible by j for $j = 2, 3, \ldots, n+1$ and it is larger than j.

17. We have

    $$\frac{(m+1)(m+2)\cdots(m+n)}{n!} = \binom{m+n}{n}.$$

    This binomial coefficient is an integer. Thus, $n! \mid (m+1)(m+2)\cdots(m+n)$.

18. Suppose on the contrary that $n = ab$ for some integers a, b where $a \geqslant 3$ is odd. Then

    $$m^n + 1 = m^{ab} + 1 = (m^b)^a + 1 = (m^b + 1)(m^{(a-1)b} - m^{(a-2)b} + \cdots + 1).$$

    As $a \geqslant 3$, both factors are larger than 1. This shows $m^n + 1$ cannot be a prime. Thus, n cannot have any odd divisor greater than 1. This means n is a power of 2.

19. Each term of the sequence can be represented as

    $$1 + 10^2 + 10^4 + \cdots + 10^{2k}$$

    for some integer $k \geqslant 1$. This is equal to

    $$\frac{10^{2(k+1)} - 1}{10^2 - 1} = \frac{(10^{k+1} - 1)(10^{k+1} + 1)}{99}.$$

    For $k \geqslant 2$, each of $10^{k+1} - 1$ and $10^{k+1} + 1$ is larger than 99. When their product is divided by 99, we still obtain a product of two integers larger than 1. Hence, it cannot be a prime number. For $k = 1$, the number 101 is a prime. Therefore, there is only one prime in the sequence.

20. Note that

$$(4n+1, kn+1) = (4n+1, (k-4)n) = (4n+1, k-4)$$

since $(4n+1, n) = 1$.

If $k-4 = \pm 2^m$ for some nonnegative integer m, then $(4n+1, k-4) = 1$ as $4n+1$ is odd.

If $k-4$ has an odd prime divisor p, then there exists an integer n' such that

$$4n'+1 \equiv 0 \pmod{p}$$

since $(4, p) = 1$. Hence, by taking $n = n'$, we have $p \mid (4n'+1, k-4)$. The condition is not satisfied in that case.

Therefore, the solutions are $k = 4 \pm 2^m$ for some nonnegative integer m.

21. WLOG assume $m \geqslant n$. We have $m^n n^m = (mn)^n n^{m-n}$. Since $m-n$ is even, n^{m-n} is a perfect square. Therefore, $(mn)^n$ is a perfect square. As n is odd, mn itself must be a perfect square.

22. Let $(m, n) = d$, and let $m = da$, $n = db$ so that $(a, b) = 1$. Then

$$\frac{m}{n} + \frac{n}{m} = \frac{a}{b} + \frac{b}{a} = \frac{a^2+b^2}{ab}.$$

We need $ab \mid a^2 + b^2$. This implies $a \mid a^2 + b^2$ so that $a \mid b^2$. As $(a, b) = 1$, this yields $a = 1$. Due to symmetry, we have $b = 1$. In that case, $\dfrac{a^2+b^2}{ab} = 2$ is an integer.

Thus, the solutions are $m = n$.

23. Let $(m, n) = d$, and let $m = da$, $n = db$ so that $(a, b) = 1$. Then

$$4d^2a^2 - d^2b^2 \mid 2d^2a^2 - d^2b^2.$$

This implies $4a^2 - b^2 \mid 2a^2 - b^2$. Thus,

$$4a^2 - b^2 \mid (4a^2 - b^2) - (2a^2 - b^2) = 2a^2.$$

Note that $(4a^2 - b^2, a^2) = (-b^2, a^2) = 1$ as $(a, b) = 1$. Therefore, we have $4a^2 - b^2 \mid 2$.

If $4a^2 - b^2 = \pm 2$, then b is even. However, this shows $4a^2 - b^2$ is a multiple of 4, which is a contradiction.

If $4a^2 - b^2 = \pm 1$, then $(2a-b)(2a+b) = \pm 1$. This is impossible since $2a + b \geqslant 3$.

Thus, there is no solution.

24. When $n = 1$, the only such integer is 7, which is a prime. When $n = 2$, both 17 and 71 are primes. We claim that all other n's are not possible.

    If $3 \mid n$, then all numbers of the desired form are multiples of 3 since $(n-1)\cdot 1+7 = n+6$ is divisible by 3.

    If $n \equiv 1 \pmod 6$ and $n \geqslant 7$, then $7 \mid 11\cdots 17$ since $7 \mid 111111$.

    If $n \equiv 2 \pmod 6$ and $n \geqslant 8$, then $7 \mid 11\cdots 171111$ since $7 \mid 111111$ and $7 \mid 60011$.

    If $n \equiv 4 \pmod 6$ and $n \geqslant 10$, then $7 \mid 11\cdots 1711111$ since $7 \mid 111111$ and $7 \mid 601111$. For $n = 4$, we check that $1711 = 29 \times 59$.

    If $n \equiv 5 \pmod 6$, then $7 \mid 11\cdots 1711$ since $7 \mid 111111$ and $7 \mid 11711$.

    Therefore, for any $n \geqslant 3$, there exists a number of the desired form which is not a prime number. Hence, n can only be 1 or 2.

25. We claim that S is any set of the form $\{kd : k \in \mathbb{Z}\}$ where d is a nonnegative integer.

    Firstly, suppose $S = \{kd : k \in \mathbb{Z}\}$. For any $ad, bd \in S$, we have $ad - bd = (a-b)d$. This is a multiple of d, and hence $ad - bd \in S$. This shows any such set S satisfies the condition.

    Secondly, consider any set S with the given property. The singleton set $S = \{0\}$ is clearly a solution. Now, suppose S contains a nonzero integer. As $m \in S$ implies $0 = m - m \in S$ and $-m = 0 - m \in S$, we may assume S contains a positive integer. Let d be the smallest positive integer that S contains. Using the base case $-d \in S$, we can prove inductively that $(-k)d \in S$ for any positive integer k.

    Next, from $0 - (-kd) = kd$, we have $kd \in S$ for any positive integer k. It follows that S contains $\{kd : k \in \mathbb{Z}\}$. If S contains another element c not divisible by d, we let $c = qd + r$ where q, r are integers and $0 < r < d$. From $r = c - qd$, we have $r \in S$. However, this contradicts the assumption that d is the smallest positive element in S. Therefore, S contains only multiples of d.

    This shows S is any set of the form $\{kd : k \in \mathbb{Z}\}$ where d is a nonnegative integer.

26. (a) False. For example, if $a = 4$ and $b = c = 2$, then $a \mid bc$ but $a \nmid b$ and $a \nmid c$.
    The result holds if a is a prime.

    (b) True. Let $\dfrac{n^2}{m^2} = k$ for some $k \in \mathbb{N}$. This yields $\dfrac{n}{m} = \sqrt{k}$. As $\sqrt{k}$ is a rational number, k must be a perfect square. This shows $\sqrt{k} \in \mathbb{N}$ and hence $m \mid n$.

    (c) True. From $mn \mid m+n$, we have $m \mid m+n$. This yields $m \mid n$. Similarly, $n \mid m$. As m and n are positive integers, these imply $m = n$.

    (d) False. For example, if $m = 4$ and $n = 12$, then $m + n = 16 \mid 48 = mn$.

    (e) True. Let $Q(x) = f(x)P(x) + r(x)$ where $f(x), r(x)$ has rational coefficients and $\deg r < \deg P$. It is given that the remainder $r(x)$ is nonzero. Suppose

$$P(n) \mid Q(n) = f(n)P(n) + r(n).$$

Let D be a positive integer such that $Df(x)$ and $Dr(x)$ have integer coefficients. The above implies
$$P(n) \mid Df(n)P(n) + Dr(n).$$
Thus, $P(n) \mid Dr(n)$. This yields $r(n) = 0$ or $|P(n)| \leqslant |Dr(n)|$. The former case has finitely many roots in n. For the latter case, as $\deg r < \deg P$, it also has finitely many solutions in n. Therefore, there are finitely many positive integers n such that $P(n) \mid Q(n)$.

(f) True. Let $m = p_1^{a_1} \cdots p_t^{a_t}$ and $n = p_1^{b_1} \cdots p_t^{b_t}$ where $a_1, \ldots, a_t, b_1, \ldots, b_t$ are nonnegative integers and $p_1, \ldots, p_t$ are distinct primes. Then
$$(m^2, n^2) = (p_1^{2a_1} \cdots p_t^{2a_t}, p_1^{2b_1} \cdots p_t^{2b_t}) = \prod_{j=1}^{t} p_j^{\min\{2a_j, 2b_j\}}$$
and
$$(m, n) = (p_1^{a_1} \cdots p_t^{a_t}, p_1^{b_1} \cdots p_t^{b_t}) = \prod_{j=1}^{t} p_j^{\min\{a_j, b_j\}}.$$
It is now obvious that $(m^2, n^2) = (m, n)^2$.

(g) False. For example, if $a = 6$, $b = 10$ and $c = 15$, then $(a, b) = 2$, $(b, c) = 5$ and $(c, a) = 3$, while $(a, b, c) = 1$.

(h) True. We always have $(m, n)[m, n] = mn$. If $[m, n] = mn$, then $(m, n) = 1$.

Prime Number

27. Suppose on the contrary that there are only finitely many prime numbers. Let these primes be
$$p_1 < p_2 < \cdots < p_k.$$
Consider the number $n = p_1 p_2 \cdots p_k + 1$. Clearly, $n > 1$. Also, we have $n \equiv 1 \pmod{p_j}$ for each p_j. This shows n is not divisible by any prime p_j. This is a contradiction since every positive integer larger than 1 is a product of primes.

28. Since $p \mid n^2$ and p is a prime, we have $p \mid n$. (Recall that $p \mid ab$ implies $p \mid a$ or $p \mid b$.) By multiplying $p \mid n$ to itself, we get $p^2 \mid n^2$.

29. Note that
$$p^4 - 1 = (p^2 - 1)(p^2 + 1) = (p - 1)(p + 1)(p^2 + 1).$$
As $p > 2$, p is odd. Since $p-1$ and $p+1$ are two consecutive even numbers, one of them is a multiple of 4 and the other is even. Also $p^2 + 1$ is even. Thus, $16 \mid p^4 - 1$.

As $p > 3$, p is of the form $3k \pm 1$. In any case, $p - 1$ or $p + 1$ is a multiple of 3.

As $p > 5$, p is not divisible by 5. If p is of the form $5k \pm 1$, then $p-1$ or $p+1$ is divisible by 5. If p is of the form $5k \pm 2$, then $p^2 + 1 \equiv (\pm 2)^2 + 1 = 4 + 1 \equiv 0 \pmod{5}$. Thus, $5 \mid p^4 - 1$ in any case.

Therefore, $16 \times 3 \times 5 = 240$ divides $p^4 - 1$.

30. Clearly, $a + b + c + d > 1$. Suppose on the contrary that $a + b + c + d = p$ is a prime. Note that

$$a + b + c + d = a + b + c + \frac{ab}{c} = \frac{ac + bc + c^2 + ab}{c} = \frac{(a + c)(b + c)}{c}.$$

This means $pc = (a + c)(b + c)$. As p is a prime, we have $p \mid a + c$ or $p \mid b + c$. WLOG assume $p \mid a + c$. Let $a + c = pn$ for some $n \in \mathbb{N}$. Then $c = n(b + c)$. This is impossible since $n(b + c) \geqslant b + c > c$.

Therefore, $a + b + c + d$ is a composite number.

31. We have

$$\begin{aligned}
& 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{1318} + \frac{1}{1319} \\
&= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{1318} + \frac{1}{1319}\right) - 2\left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{1318}\right) \\
&= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{1318} + \frac{1}{1319}\right) - \left(1 + \frac{1}{2} + \cdots + \frac{1}{659}\right) \\
&= \frac{1}{660} + \frac{1}{661} + \cdots + \frac{1}{1319} \\
&= \left(\frac{1}{660} + \frac{1}{1319}\right) + \left(\frac{1}{661} + \frac{1}{1318}\right) + \cdots + \left(\frac{1}{989} + \frac{1}{990}\right) \\
&= 1979 \cdot \left(\frac{1}{660 \cdot 1319} + \frac{1}{661 \cdot 1318} + \cdots + \frac{1}{989 \cdot 990}\right).
\end{aligned}$$

One may check that 1979 is a prime. Thus, this factor cannot be cancelled out by the denominator. Thus, p must be divisible by 1979.

32. Note that if a prime p divides one of the numbers, then p has to divide another number. Otherwise, only the product of numbers in one subset is a multiple of p. Since there are only 6 consecutive numbers, a prime number p can divide two of these numbers only if $p \leqslant 5$.

Also, the prime number 5 must divide one, and hence two, of the numbers. The only possibility is that $5 \mid n$ and $5 \mid n + 5$. Thus, each of $n + 1, n + 2, n + 3, n + 4$ only has the prime divisors 2 and/or 3. As two of them are odd numbers, either $n + 1, n + 3$ or $n + 2, n + 4$, these two numbers must be a power of 3. However, their difference is 2 so that one of them cannot be a multiple of 3. This contradicts the fact that they are integers larger than 1.

Thus, there is no such set.

33. The product of the first n natural numbers is $n!$, while their sum is

$$1+2+\cdots+n=\frac{n(n+1)}{2}.$$

If $n+1$ is an odd prime, then $n+1 \mid \dfrac{n(n+1)}{2}$. However, we have $n+1 \nmid n!$ since $n+1$ is a prime (otherwise $n+1$ has to divide one of $1,2,\ldots,n$).

If $n+1$ is not an odd prime, then either $n+1=2$ or $n+1$ is a composite number. The former case $n=1$ is trivial as $1 \mid 1$.

Suppose $n+1$ is a composite number. We shall prove that $\dfrac{n(n+1)}{2} \mid n!$. This is the same as

$$n+1 \mid 2(n-1)!.$$

If $n+1=4$, then this is true since $4 \mid 2\times 3! = 12$. If $n+1=p^2$ where p is an odd prime, then $2p \leqslant n-1$. Thus, $(n-1)!$ is divisible by $n+1=p^2$.

If $n+1$ is composite but is not the square of a prime, then we can find distinct positive integers $a,b \geqslant 2$ such that $n+1=ab$. In that case, we have $n-1=ab-2 \geqslant 2a-2 \geqslant a$. Similarly, $n-1 \geqslant b$. As a,b are distinct, we have $n+1=ab \mid (n-1)!$.

Therefore, if $n+1$ is not an odd prime, then $\dfrac{n(n+1)}{2} \mid n!$. The equivalence is established.

34. Note that

$$c_{m+j}-c_k=(m+j)(m+j+1)-k(k+1)=(m+j-k)(m+j+k+1)$$

for $j=1,2,\ldots,n$. This means we have to prove

$$\prod_{j=1}^{n}(m+j-k)(m+j+k+1)=\frac{(m+n-k)!(m+n+k+1)!}{(m-k)!(m+k+1)!}$$

is divisible by $c_1c_2\cdots c_n = 1\cdot 2\cdot 2\cdot 3\cdots n(n+1) = n!(n+1)!$. Indeed, we have

$$\begin{aligned}&\frac{(c_{m+1}-c_k)(c_{m+2}-c_k)\cdots(c_{m+n}-c_k)}{c_1c_2\cdots c_n}\\ =\;&\frac{(m+n-k)!(m+n+k+1)!}{(m-k)!(m+k+1)!n!(n+1)!}\\ =\;&\frac{1}{m+k+1}\binom{m+n+k+1}{n+1}\binom{m+n-k}{n}.\end{aligned}$$

As the binomial coefficient is an integer, it suffices to show the prime $p=m+k+1$

divides $\binom{m+n+k+1}{n+1}$. Note that

$$\begin{aligned}\binom{m+n+k+1}{n+1} &= \frac{(m+n+k+1)(m+n+k)\cdots(m+k+1)}{(n+1)!} \\ &= \frac{(m+n+k+1)(m+n+k)\cdots(m+k+2)p}{(n+1)!}.\end{aligned}$$

Since this is an integer and p is a prime greater than each term in the denominator, this integer is divisible by p. This completes the proof.

35. WLOG assume $p \leqslant q$. We have $q^2 \mid p^3+1 = (p+1)(p^2-p+1)$. Note that

$$(p+1, p^2-p+1) = (p+1, (p-2)(p+1)+3) = (p+1, 3).$$

If $q \mid p+1$ and $q \mid p^2-p+1$, then we have $q \mid 3$. The only possibility is $q=3$. In that case, we need $p^2 \mid q^3+1 = 28$. The only possibility is $p=2$. We check that $(p,q)=(2,3)$ is a solution.

If q divides exactly one of $p+1$ and p^2-p+1, then we have $q^2 \mid p+1$ or $q^2 \mid p^2-p+1$. Since $q^2 \geqslant p^2 > p+1$ for $p \geqslant 2$, the former case is rejected. For $q^2 \mid p^2-p+1$, we have

$$p^2-p+1 \geqslant q^2 \geqslant p^2,$$

which is again impossible.

Therefore, the only solutions are $(p,q)=(2,3),(3,2)$.

36. Note that

$$(m+n)^3+8 = (m+n)^3+2^3 = (m+n+2)((m+n)^2-2(m+n)+4).$$

Since p divides this product, we have $p \mid m+n+2$ or $p \mid (m+n)^2-2(m+n)+4$.

If $p \mid m+n+2$, then $m^2+n^2 \leqslant m+n+2$. For $m=1$, we have $n^2 \leqslant n+2$. This yields $n=1,2$. Thus, $p=2,5$. We check that both are solutions. For $m \geqslant 2$, we have $m^2 \geqslant m+2$ so that $n^2 \leqslant n$. The only possibility is $n=1$, which is symmetric to the case $m=1$.

If $p \mid (m+n)^2-2(m+n)+4$, then $m^2+n^2 \mid 2mn-2m-2n+4$. Since we may assume p is odd (as the case $p=2$ has been solved), we can remove the factor 2 on the right. This gives $m^2+n^2 \mid mn-m-n+2$. Thus

$$m^2+n^2 \leqslant mn-m-n+2 \leqslant \frac{m^2+n^2}{2}-m-n+2.$$

Therefore, we have $m^2+n^2<4$. There is no solution for odd prime p.

Hence, p can be 2 or 5.

<u>Size</u>

37. Note that $3n-1$ and $2n+2$ are positive integers. Therefore, we have $3n-1 \leqslant 2n+2$. This yields $n \leqslant 3$. By checking $n = 1, 2, 3$, we find that only $n = 1, 3$ are solutions.

38. Since n^2+1 and $2n+1$ are positive integers, we have $n^2+1 \leqslant 2n+1$. This implies $n^2 \leqslant 2n$. The only possibilities are $n = 1, 2$. We check that only $n = 2$ is a solution.

39. From $2n^2+3 \mid n^3+3n+6$, we have $2n^2+3 \mid 2(n^3+3n+6) = n(2n^2+3)+3n+12$. This yields $2n^2+3 \mid 3n+12$. As $2n^2+3$ and $3n+12$ are positive integers, this implies $2n^2+3 \leqslant 3n+12$, i.e. $2n^2-3n-9 \leqslant 0$. It follows that

$$n \leqslant \frac{3+\sqrt{(-3)^2-4(2)(-9)}}{2(2)} = 3.$$

By checking $n = 1, 2, 3$, we find that both $n = 1, 3$ are solutions.

40. From the divisibility, we have $3n-12 = 0$ or $n^2+2 \leqslant |3n-12|$.

Clearly, $n = 4$ is a solution. By checking $n = 1, 2, 3$, we find that both $n = 1, 2$ are solutions.

For $n \geqslant 5$, we have $n^2+2 \leqslant 3n-12$. This yields $n^2-3n+14 \leqslant 0$. This is impossible since $\Delta = (-3)^2-4(1)(14) < 0$.

Hence, the only solutions are $n = 1, 2, 4$.

41. Since mn and $m+n$ are positive integers, we have $mn \leqslant m+n$. This yields

$$(m-1)(n-1) \leqslant 1.$$

If $m = 1$, then we need $n \mid n+1$, i.e. $n \mid 1$. The only solution is $(m, n) = (1, 1)$. Similarly, we get the same solution if $n = 1$.

If $m, n \geqslant 2$, then $m-1, n-1 \geqslant 1$. Thus, the only possibility is $m = n = 2$. We check that this is another solution.

Hence, the solutions are $(m, n) = (1, 1), (2, 2)$.

42. WLOG assume $m \geqslant n$. If $m = n$, then from $m \mid n+1$ we have $m \mid 1$. The only possibility is $m = n = 1$, which is a solution by simple checking.

If $m \geqslant n+1$, then from $m \mid n+1$ we have $m \leqslant n+1$. Together with $m \geqslant n+1$, we have $m = n+1$. The other divisibility becomes $n \mid n+2$, i.e. $n \mid 2$. The solutions are $(m, n) = (2, 1), (3, 2)$.

Hence, the solutions are $(m, n) = (1, 1), (2, 1), (1, 2), (3, 2), (2, 3)$.

43. From $abc-1 \mid ab^2$, we have $abc-1 \mid ab^2c = b(abc-1)+b$. Hence, $abc-1 \mid b$. As $abc-1$ is a positive integer, this gives $abc-1 \leqslant b$. If $ac \geqslant 2$, then $2b-1 \leqslant b$ so that $b \leqslant 1$. The only possibility is $b = 1$. Then we have $ac-1 \mid 1$. This yields $ac-1 = 1$. The solutions are $(a,b,c) = (1,1,2),(2,1,1)$. Clearly, these are solutions.

If $ac = 1$, then $a = c = 1$ and $b-1 \mid b^2$. So $b-1 \mid 1$, which yields $b-1 = 1$. This gives another solution $(a,b,c) = (1,2,1)$.

Thus, the solutions are $(a,b,c) = (1,1,2),(2,1,1),(1,2,1)$.

44. Firstly, from $m \mid 2n+1$, we know that m is odd. Then $n \mid 3m+2$ implies n is odd. Taking the product of the divisibilities, we have

$$mn \mid (2n+1)(3m+2) = 6mn + 3m + 4n + 2.$$

This yields $mn \mid 3m+4n+2$ and hence

$$mn \leqslant 3m + 4n + 2.$$

This can be rewritten as

$$(m-4)(n-3) \leqslant 14.$$

If $n = 1$, then $m \mid 2n+1 = 3$. The only solutions are $(m,n) = (1,1),(3,1)$. Note that $n \mid 3m+2$ must hold.

If $n = 3$, then we cannot have $n \mid 3m+2$.

If $n = 5$, then $m \mid 2n+1 = 11$. The only possible solutions are $(m,n) = (1,5),(11,5)$. In both cases, the other divisibility $n \mid 3m+2$ holds.

If $n \geqslant 7$ and $m = 1$, then $n \mid 3m+2 = 5$. This is impossible as $n \geqslant 7$.

If $n \geqslant 7$ and $m = 3$, then $n \mid 3m+2 = 11$. The only possible solution is $(m,n) = (3,11)$, which is not a solution by checking.

If $n \geqslant 7$ and $m = 5$, then $n \mid 3m+2 = 17$. The only possible solution is $(m,n) = (5,17)$, which is indeed a solution by checking.

If $m,n \geqslant 7$, then the only possibility for which $(m-4)(n-3) \leqslant 14$ is $(m,n) = (7,7)$. However, we check that this is not a solution.

Therefore, the only solutions are $(m,n) = (1,1),(3,1),(1,5),(11,5),(5,17)$.

45. From $m^2-n^2 \mid m+4n$ and $m^2-n^2 = (m-n)(m+n)$, we have $m+n \mid m+4n$. Since

$$m+n < m+4n < 4(m+n),$$

we can only have $2(m+n) = m+4n$ or $3(m+n) = m+4n$.

If $2(m+n) = m+4n$, then $m = 2n$. The divisibility becomes $3n^2 \mid 6n$, i.e. $n \mid 2$. The solutions are $(m,n) = (2,1),(4,2)$.

If $3(m+n) = m+4n$, then $2m = n$. The divisibility becomes $-3m^2 \mid 9m$, i.e. $m \mid 3$. The solutions are $(m,n) = (1,2),(3,6)$.

Therefore, the solutions are $(m,n) = (2,1),(4,2),(1,2),(3,6)$.

46. If $a \geqslant 4$, then $b \geqslant 5$ and $c \geqslant 6$. Then we have

$$\frac{abc-1}{(a-1)(b-1)(c-1)} < \frac{a}{a-1} \cdot \frac{b}{b-1} \cdot \frac{c}{c-1} \leqslant \frac{4}{3} \cdot \frac{5}{4} \cdot \frac{6}{5} = 2.$$

This implies $abc - 1 = (a-1)(b-1)(c-1)$. This is the same as

$$ab + bc + ca = a + b + c.$$

However, this is impossible since $ab > a$, $bc > b$ and $ca > c$.

If $a = 3$, then we need $2(b-1)(c-1) \mid 3bc - 1$. Similar to the above steps, we have

$$1 < \frac{3bc-1}{2(b-1)(c-1)} < \frac{3}{2} \cdot \frac{b}{b-1} \cdot \frac{c}{c-1} \leqslant \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} = \frac{5}{2}.$$

The only possibility is $\frac{3bc-1}{2(b-1)(c-1)} = 2$. This is the same as $bc - 4b - 4c + 5 = 0$, i.e.

$$(b-4)(c-4) = 11.$$

Thus, $b - 4 = 1$ and $c - 4 = 11$. This gives the solution $(a, b, c) = (3, 5, 15)$.

If $a = 2$, then we need $(b-1)(c-1) \mid 2bc - 1$. As $(b-1)(c-1) = bc - b - c + 1$, we have

$$2bc - 1 \equiv 2(b + c - 1) - 1 = 2b + 2c - 3 \pmod{(b-1)(c-1)}.$$

Thus, $(b-1)(c-1) \mid 2b + 2c - 3$. It follows that

$$(b-1)(c-1) \leqslant 2b + 2c - 3,$$

i.e.

$$bc - 3b - 3c + 4 \leqslant 0.$$

This is the same as

$$(b-3)(c-3) \leqslant 5.$$

If $b = 3$, then we need $2(c-1) \mid 2c + 3$. This is impossible since $2c + 3$ is odd. If $b = 4$, then we need $3(c-1) \mid 2c + 5$. In particular, we have $c - 1 \mid 2c + 5$ so that $c - 1 \mid 7$. The only possibility is $c = 8$. One checks that $(a, b, c) = (2, 4, 8)$ is a solution. If $b \geqslant 5$, then $c - 3 \leqslant 2$ so that $c \leqslant 5 \leqslant b$. This is impossible.

Hence, the solutions are $(a, b, c) = (3, 5, 15), (2, 4, 8)$.

47. If two of a, b, c are equal, say $a = b$, then $a \mid bc - 1$ implies $a \mid -1$. Thus, we have $a = b = 1$. In that case, all divisibilities are satisfied.

Suppose a, b, c are pairwise distinct. If $a = 1$, then we need $b \mid c - 1$ and $c \mid b - 1$. WLOG assume $b < c$. Then we cannot have $c \mid b - 1$.

WLOG we may assume $a > b > c > 1$. Taking the product of the divisibilities, we get

$$abc \mid (bc-1)(ca-1)(ab-1) = a^2b^2c^2 - a^2bc - ab^2c - abc^2 + ab + bc + ca - 1.$$

Hence, $abc \mid ab + bc + ca - 1$. This implies

$$abc \leqslant ab + bc + ca - 1.$$

Note that $ab - a - b = (a-1)(b-1) - 1 > 0$. Therefore, we have

$$ab - 1 \geqslant c(ab - a - b) \geqslant 2(ab - a - b).$$

This yields $ab \leqslant 2a + 2b - 1$, i.e.

$$(a-2)(b-2) \leqslant 3.$$

Since $b \geqslant 3$ and $a \geqslant 4$, the only possibilities are $(a, b) = (4, 3), (5, 3)$.

If $a = 4$ and $b = 3$, then $c \mid ab - 1 = 11$. There is no solution for $b > c > 1$.

If $a = 5$ and $b = 3$, then equality holds. This shows $c = 2$. We check that this is a solution.

Therefore, the solutions are $(a, b, c) = (1, 1, t), (2, 3, 5)$ and their permutations.

48. From $b - k \mid a - k^n$ and $k \equiv b \pmod{b-k}$, we have $b - k \mid a - b^n$. Since the right-hand side is a constant and this holds for any k, we must have $a - b^n = 0$, i.e. $a = b^n$.

49. Let $a = m^2 + \alpha$, $b = m^2 + \beta$ and $d = m^2 + \delta$. Then $1 \leqslant \alpha, \beta, \delta \leqslant m - 1$ and they are integers. The condition $d \mid ab$ is the same as $m^2 + \delta \mid (m^2 + \alpha)(m^2 + \beta)$. Note that $m^2 \equiv -\delta \pmod{m^2 + \delta}$. Therefore, we can replace m^2 by $-\delta$. This gives

$$m^2 + \delta \mid (\alpha - \delta)(\beta - \delta).$$

If $\delta = \alpha, \beta$, then $d = a, b$, which is clearly a solution.

If $\delta \neq \alpha, \beta$, then we have $m^2 + \delta \leqslant |(\alpha - \delta)(\beta - \delta)|$. As α, β, δ are integers lying between 1 and $m - 1$, their differences are at most $m - 2$. Therefore,

$$m^2 + \delta \leqslant |(\alpha - \delta)(\beta - \delta)| \leqslant (m-2)^2.$$

Noting that $\delta > 0$, this yields $m^2 < (m-2)^2$. This is impossible for $m = 1$. For $m \geqslant 2$, this is the same as $m < m - 2$, which is again impossible.

Thus, the only solutions are $d = a$ and $d = b$.

50. Firstly, from

$$p \mid n^3 - 1 = (n-1)(n^2 + n + 1),$$

we have $p \mid n - 1$ or $p \mid n^2 + n + 1$ since p is a prime. As $n \mid p - 1$ suggests $n < p$, the former case $p \mid n - 1$ is rejected. Hence, $p \mid n^2 + n + 1$.

Let $p - 1 = an$ and $n^2 + n + 1 = bp$. These combine to give

$$n^2 + n + 1 = b(an + 1).$$

Consider modulo n. Then $1 \equiv b \pmod{n}$. If $b = 1$, then $p = n^2 + n + 1$, and hence

$$4p - 3 = 4n^2 + 4n + 1 = (2n + 1)^2.$$

If $b > 1$, then $b \geqslant n + 1$ as $b \equiv 1 \pmod{n}$. It follows that

$$n^2 + n + 1 = b(an + 1) \geqslant (n + 1)(n + 1) = n^2 + 2n + 1.$$

This is a contradiction.

Thus, only the case $b = 1$ can hold, and so $4p - 3$ is a perfect square.

51. From $mn - 1 \mid m^2 - n$, we have $mn - 1 \mid (m^2 - n)n = m(mn - 1) + m - n^2$. This implies $mn - 1 \mid n^2 - m$. This is symmetric to the given divisibility. Therefore, we may assume $m \leqslant n$.

    From $mn - 1 \mid m^2 - n$, we get $m^2 - n = 0$ or $mn - 1 \leqslant |m^2 - n|$. The former case $n = m^2$ is clearly a solution. For the latter case, if $m^2 - n > 0$, then $mn - 1 \leqslant m^2 - n$. This implies

    $$m^2 - n \geqslant mn - 1 > mn - n$$

    so that $m > n$. This contradicts $m \leqslant n$.

    If $m^2 - n < 0$, let $n = m^2 + k$ where $k \geqslant 1$. Then $mn - 1 \leqslant |m^2 - n|$ becomes $m(m^2 + k) - 1 \leqslant k$. This implies

    $$k \geqslant m^3 + mk - 1 > mk > k,$$

    which is impossible.

    Hence, the only solutions are $(m, n) = (t, t^2), (t^2, t)$ for some integer $t > 1$.

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52. Clearly, $(a, b, c) = (0, 0, 0)$ is a solution. Suppose there exists a solution for which a, b, c are not all zeros. We pick a solution such that $|a| + |b| + |c|$ is minimized.

    Note that a is even. Let $a = 2m$. Then the equation becomes $4m^3 + b^3 + 2c^3 = 0$. Similarly, as b is even, we let $b = 2n$. The equation becomes $2m^3 + 4n^3 + c^3 = 0$. This shows (c, m, n) is another solution to the original equation. Note that

    $$|c| + |m| + |n| = |c| + \left|\frac{a}{2}\right| + \left|\frac{b}{2}\right| \leqslant |c| + |a| + |b|.$$

    By the minimality choice of the solution, either $(c, m, n) = (0, 0, 0)$ or equality holds. The former case yields the contradiction $(a, b, c) = (0, 0, 0)$. The latter case only holds when $a = b = 0$. However, this also implies $c = 0$ from the equation.

    Thus, there is no solution except $(a, b, c) = (0, 0, 0)$.

53. Suppose such a triple exists. As $a+2b>2$ is a power of 2, a must be even. Let $a=2m$.

As $c+2a=c+4m>4$ is a power of 2, c must be a multiple of 4.

Inductively, we can prove that for each $k\in\mathbb{N}$, one of the variables a,b,c is a multiple of 2^k. This is impossible as a,b,c are fixed positive integers.

54. Suppose the positive integers are $(x_1,x_2,\ldots,x_{2k+1})$. Let $S=x_1+x_2+\cdots+x_{2k+1}$. By removing any x_j, the sum of the remaining numbers is $S-x_j$. As the numbers can be partitioned into two groups with the same sum, $S-x_j$ must be even. Thus, $S-x_i\equiv S-x_j \pmod 2$ for any i,j. This shows all x_j's have the same parity.

If all the x_j's are even, then the positive integers $\left(\frac{x_1}{2},\frac{x_2}{2},\ldots,\frac{x_{2k+1}}{2}\right)$ also satisfy the condition. Indeed, all the sums concerned are divided by 2 so that the sums of numbers of the two groups remain equal for any partition.

If all the x_j's are odd, then the positive integers $\left(\frac{x_1+1}{2},\frac{x_2+1}{2},\ldots,\frac{x_{2k+1}+1}{2}\right)$ also satisfy the condition. Indeed, all the sums concerned are added by k and then divided by 2 so that the sums of numbers of the two groups remain equal for any partition.

Suppose on the contrary that the positive integers are not all equal. We carry out the above operations to generate new sets of positive integers satisfying the condition. Note that the numbers in the new sets are not equal as well. Also, as $\frac{x}{2}<x$ for even x and $\frac{x+1}{2}<x$ for odd $x\geqslant 3$, the sums S of the new sets are strictly decreasing (note that not all the integers can be equal to 1). This creates a strictly decreasing sequence of positive integers, which is impossible.

Therefore, all the integers are equal.

55. If $a=b$, then
$$\frac{b+1}{a}+\frac{a+1}{b}=\frac{2(a+1)}{a}=2+\frac{2}{a}.$$
This is an integer if and only if $a=1,2$. When $a=b=1$, $k=4$. When $a=b=2$, $k=3$.

Suppose $k\neq 3,4$. We claim that there are no positive integer solutions to
$$a^2+b^2+a+b=kab. \tag{1}$$
Suppose this is solvable. Choose a positive integer solution to (1) for which $a+b$ is minimized. WLOG assume $a>b$. This can be rewritten as
$$a^2+(1-kb)a+(b^2+b)=0.$$
As a quadratic equation in a, there is another solution a'. By Vieta's theorem, we have
$$\begin{aligned}a+a'&=kb-1,\\ aa'&=b^2+b.\end{aligned}$$

From the first equation, a' is an integer. From the second equation, a' is positive, and we have

$$a' = \frac{b^2+b}{a} \leqslant \frac{(a-1)^2+(a-1)}{a} = a-1.$$

This shows $a' < a$. This gives a pair (a', b) of positive integers with $\dfrac{b+1}{a'} + \dfrac{a'+1}{b} = k$ and $a'+b < a+b$. This contradicts the minimality of $a+b$.

Therefore, k can only be 3 or 4.

56. Let $\dfrac{m^2+n^2+6}{mn} = k$. For fixed k, we choose a positive integer solution for which $m+n$ is minimized. WLOG assume $m \geqslant n$. The equation can be rewritten as

$$m^2 - knm + (n^2+6) = 0.$$

As a quadratic equation in m, there is another solution m'. By Vieta's theorem, we have

$$\begin{aligned} m + m' &= kn, \\ mm' &= n^2+6. \end{aligned}$$

From the first equation, m' is an integer. From the second equation, m' is positive. Note that the pair (m', n) satisfies

$$\frac{(m')^2+n^2+6}{m'n} = k$$

by reversing the above steps. From the minimality of $m+n$, we have $m' \geqslant m$. Then

$$n^2+6 = mm' \geqslant m^2.$$

Recall that we have assumed $m \geqslant n$. If $m = n$, then the given divisibility becomes $m^2 \mid 2m^2+6$, i.e. $m^2 \mid 6$. The only solution is $m = n = 1$. This gives $k = 8$.

If $m \geqslant n+1$, then $n^2+6 \geqslant m^2 \geqslant (n+1)^2$. This yields $n \leqslant 2$.

For $n = 2$, as $m^2 \leqslant n^2+6 = 10$ and $m \geqslant n+1 = 3$, it suffices to check $m = 3$. The divisibility is not satisfied.

For $n = 1$, as $m^2 \leqslant n^2+6 = 7$ and $m \geqslant n+1 = 2$, it suffices to check $m = 2$. The divisibility is not satisfied.

Therefore, k can only be 8.

57. Suppose $\dfrac{a^2+b^2}{ab+1} = k$ where k is not a perfect square. For this fixed k, we choose a positive integer solution (a, b) for which $a+b$ is the minimal. WLOG assume $b \leqslant a$.

The equation can be rewritten as

$$a^2 - kba + (b^2 - k) = 0.$$

This is a quadratic equation in a. Let a' be the other root of the equation. Then

$$\begin{aligned} a + a' &= kb, \\ aa' &= b^2 - k. \end{aligned}$$

Clearly, $a' = kb - a$ is an integer. We claim that it is a nonnegative integer. Indeed, we have

$$\begin{aligned} & kb - a > -1 \\ \Leftrightarrow \quad & \frac{a^2b + b^3}{ab + 1} > a - 1 \\ \Leftrightarrow \quad & a^2b + b^3 > a^2b - ab + a - 1 \\ \Leftrightarrow \quad & b^3 + ab + 1 > a. \end{aligned}$$

This is obviously true. Thus, a' is a nonnegative integer. If $a' = 0$, then $b^2 - k = 0$. This contradicts the assumption that k is not a perfect square. If a' is a positive integer, then the pair (a', b) of positive integers satisfies

$$\frac{(a')^2 + b^2}{a'b + 1} = k.$$

From the minimality of the pair (a, b), we must have $a' \geqslant a$. This implies

$$b^2 - k = aa' \geqslant a^2 \geqslant b^2.$$

This is a contradiction. Thus, k has to be a perfect square.

58. Firstly, since we can take x and y to be negative without affecting the first three conditions, condition (iv) is somehow redundant. Therefore, we shall ignore this condition in the following and consider positive integers x, y.

Clearly, $(x_0, y_0) = (1, 1)$ is such a pair. Suppose we have a pair (x_k, y_k) of positive integers satisfying (i), (ii) and (iii) with $x_k + y_k > x_j + y_j$ for $j = 0, 1, \ldots, k - 1$. Due to symmetry, we may assume $x_k \leqslant y_k$. From (ii) and (iii), we have

$$x_k \mid x_k^2 + y_k^2 + m \quad \text{and} \quad y_k \mid x_k^2 + y_k^2 + m.$$

As x_k, y_k are relatively prime, these yield $x_k y_k \mid x_k^2 + y_k^2 + m$. Let

$$x_k^2 + y_k^2 + m = t x_k y_k$$

for some integer t.

The quadratic equation $x^2 - ty_kx + (y_k^2 + m) = 0$ in x has two real roots, one of which is x_k. Suppose the other root is x'. Then we have

$$\begin{aligned} x_k + x' &= ty_k, \\ x_kx' &= y_k^2 + m. \end{aligned}$$

The first relation shows that x' is an integer, while the second relation shows that it is positive. We shall show that the pair (x', y_k) satisfies the conditions.

Firstly, we have

$$(x', y_k) = (ty_k - x_k, y_k) = (-x_k, y_k) = 1.$$

By reversing the above steps, we find that $x'y_k \mid (x')^2+y_k^2+m$. This yields $x' \mid y_k^2+m$ and $y_k \mid (x')^2 + m$. Hence, conditions (i), (ii) and (iii) are satisfied. Also, from $y_k \geqslant x_k > 0$ and $m > 0$, we get

$$x' > \frac{y_k^2}{x_k} \geqslant y_k \geqslant x_k > 0$$

so that the new pair (x', y_k) has a larger sum than the pair (x_k, y_k). It follows that this pair is different from all the pairs obtained before. We define (x_{k+1}, y_{k+1}) by $x_{k+1} = y_k$ and $y_{k+1} = x'$. This process defines a recursive relation to generate infinitely many pairs of integers satisfying the conditions.